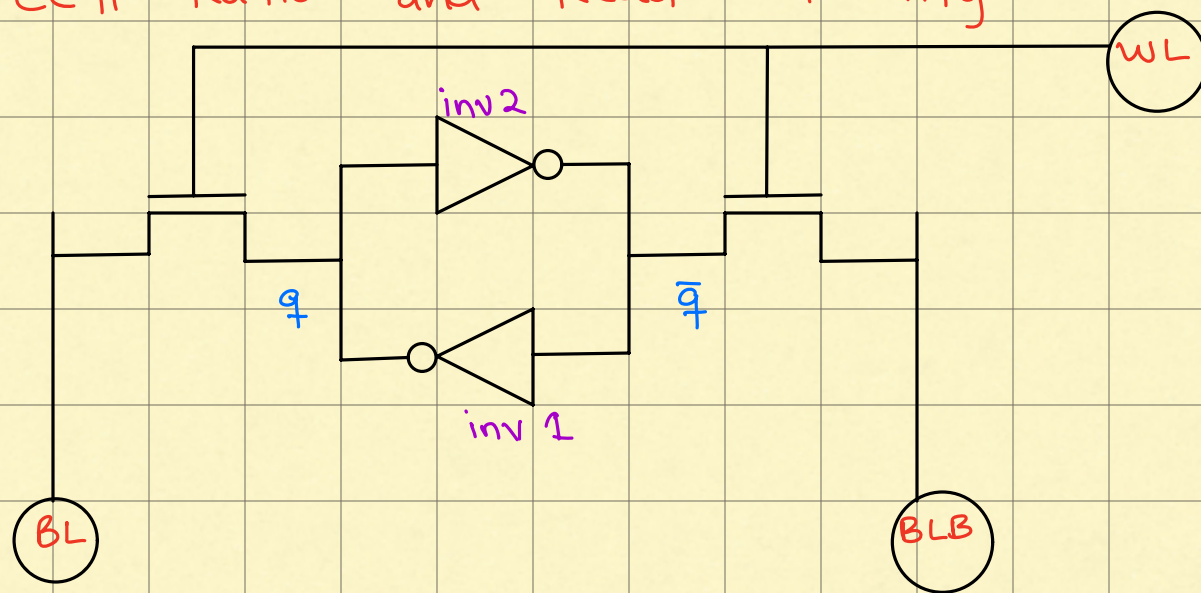


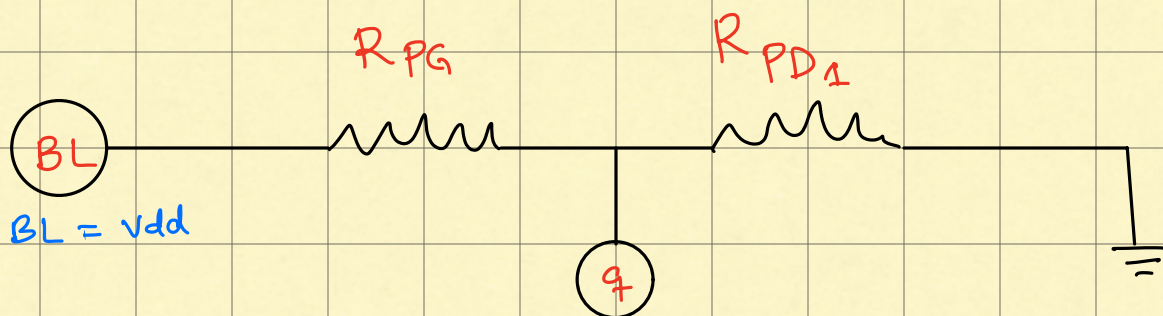
READ STABILITY ANALYSIS

1.1.1 Cell Ratio and Read Stability



For the 6T1R1B1 SRAM, assume $q = 0$ for read stability analysis. Given $q = 0$ implies it is pulled down by NMOS inside $inv1$.

The path is therefore defined by approx voltage divider model



By voltage division,
$$\frac{V_q}{V_{DD}} = \frac{R_{PD1}}{R_{PG} + R_{PD1}}$$

$$\Rightarrow \frac{V_g}{V_{dd}} = \frac{1}{1 + R_{PG} / R_{PD1}}$$

From transistor physics, $R_T \propto \frac{1}{W_T}$

WLG, assume both Pass transistor and pull down transistor all have same length

$$\text{Let } \beta = \frac{(W/L)_{PD}}{(W/L)_{PG}} = \frac{W_{PD}}{W_{PG}} \quad \text{WLG}$$

$$\Rightarrow \beta \approx \frac{R_{PG}}{R_{PD}}$$

But $\frac{V_g}{V_{dd}} \approx \frac{1}{1 + R_{PG} / R_{PD1}}$

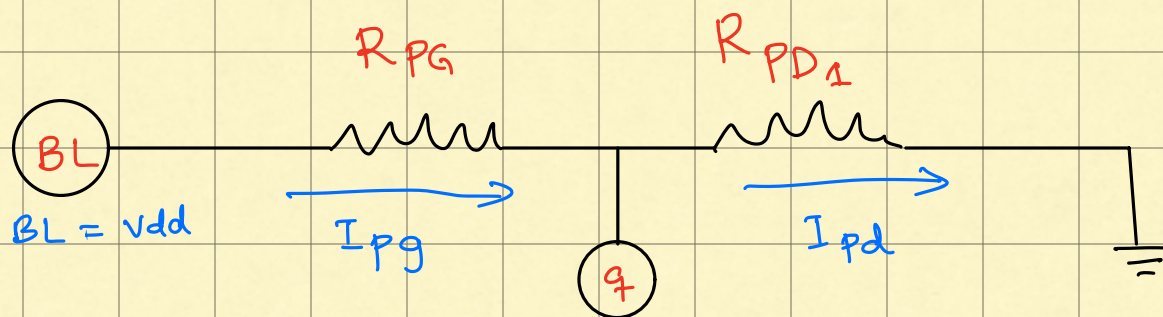
$$\Rightarrow \frac{V_g}{V_{dd}} \approx \frac{1}{1 + \beta}$$

$$\Rightarrow V_q \approx \frac{V_{dd}}{1 + \beta}$$

From approximation, it is illustrated that the value of β as selected reduces the value of V_q . The relevance is reinforced by the transition voltage of the transistor. Assume V_m where both pmos & Nmos are conducting and $V_{in}^1 = V_{out}^1$. At V_m , transistor is at transition and is not stable for a prior state voltage.

1.1.2 Mosfet current Balance Analysis.

At equilibrium read-disturb voltage, the storage-node voltage is constant. Neglecting gate leakage and other secondary currents, Kirchhoff current imply that $I_{pd} = I_{pg}$



$$|I_{pg}| = |I_{pd}|$$

For pass transistor,

$$\begin{aligned} D &= v_{dd} \\ G &= v_{dd} \\ S &= V_q \end{aligned}$$

$$\begin{aligned} \Rightarrow V_{ds} &= v_{dd} - V_q \\ V_{gs} &= v_{dd} - V_q \end{aligned}$$

By transistor physics, $V_{ds} > V_{gs} - V_{th}$ for all cases since $v_{ds} = v_{gs}$ (wLG) and pass transistor therefore operates in saturation

$$\Rightarrow I_{pg} = \frac{1}{2} k_{pg} (V_{gs} - V_{th})^2$$

$$k_{pg} = \mu_{n} C_{ox} \left(\frac{w}{L} \right)_{pg}$$

For pull-down Nmos pd1

$$V_d = V_q$$

$$V_g = \overline{V_q} \approx V_{dd}$$

$$V_s = 0$$

$$V_{ds} = V_q$$

$$V_{gs} = V_{dd}$$

$$V_{ds} < V_{gs} - V_{th}$$

thus pull down

operates in triode region.

$$I_{pd} = k_{pd} \left[(V_{gs} - V_{th}) V_{ds} - \frac{V_{ds}^2}{2} \right]$$

$$k_{pd} = \mu_n C_{ox} \left(\frac{w}{L} \right)_{pd}$$

$$I_{pd} = I_{pg}$$

for read disturb eq. voltage

$$k_{pd} \left[(V_{dd} - V_{tn}) V_g - \frac{V_g^2}{2} \right] = \frac{1}{2} k_{pg} (V_{dd} - V_g - V_{tn})^2$$

$$\frac{k_{pd}}{k_{pg}} = \frac{(V_{dd} - V_g - V_{tn})^2}{2(V_{dd} - V_{tn}) V_g - V_g^2}$$

Let $E = V_{dd} - V_{tn}$

$$\frac{k_{pd}}{k_{pg}} = \frac{(E - V_g)^2}{2E V_g - V_g^2}$$

Also, $\frac{k_{pd}}{k_{pg}} = \frac{(w/L)_{pd}}{(w/L)_{pg}} = \beta$

$$(2E V_g - V_g^2) \cdot \beta = (E - V_g)^2$$

$$2E\beta V_g - \beta V_g^2 = E^2 + V_g^2 - 2E V_g$$

$$V_g^2(1 + \beta) + V_g(-2E - 2E\beta) + E^2$$

By quadratic formula,

$$V_g = E \cdot \left(1 \pm \sqrt{\frac{\beta}{1 + \beta}} \right)$$

The +ve solution exceeds $V_{dd} - V_{tn}$ and is inconsistent with the assumed conducting pass gate and triode region pull-down.

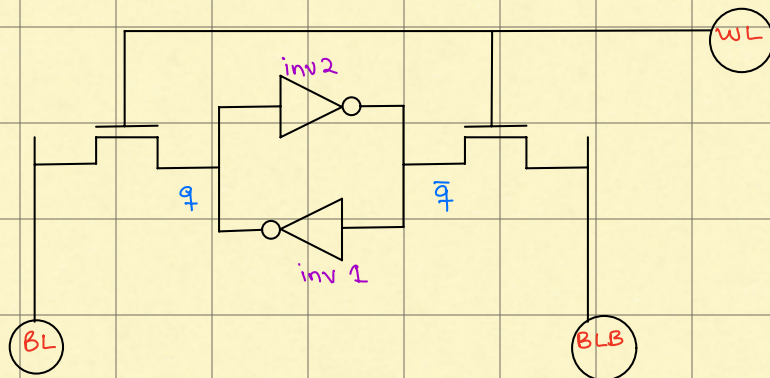
$$\Rightarrow V_q = (V_{dd} - V_{tn}) \left(1 - \sqrt{\frac{\beta}{1+\beta}} \right)$$

This result illustrates that increasing the cell ratio β reduces the voltage developed at the stored zero node during a read.

The derivation assumes ideal long channel square law behavior, equal NMOS threshold voltages, negligible body effect and $\bar{Q} \propto V_{dd}$.

1.1.3 Inverter Switching Voltage

The disturbed node q is connected to the input of inv 2



Switching voltage is defined by $V_{in} = V_{out} = V_m$

At inverter's DC switch point, the n_{mos} & p_{mos} drain current magnitudes are equal

$$I_{DN} = |I_{DP}|$$

Assuming both mosfets are in saturation and using square law current equations,

$$\frac{1}{2} k_n (V_m - V_{tn})^2 = \frac{1}{2} k_p (V_{dd} - V_m - |V_{tp}|)^2$$

$$\frac{k_p}{k_n} = \frac{(V_m - V_{tn})^2}{(V_{dd} - V_m - |V_{tp}|)^2}$$

$$\text{Let } \sqrt{\frac{k_p}{k_n}} = B$$

$$\Rightarrow B (V_{dd} - V_m - |V_{tp}|) = V_m - V_{tn}$$

$$V_m = \frac{B (V_{dd} - |V_{tp}|) + V_{tn}}{1 + B}$$

$$V_m = \frac{V_{tn} + B (V_{dd} - |V_{tp}|)}{1 + B}$$

$$\text{Also, } \frac{k_p}{k_n} = \frac{\mu_p C_{ox} (W/L)_p}{\mu_n C_{ox} (W/L)_n}$$

$$\text{And } \beta = (W/L)_p, \quad \gamma = (W/L)_n$$

$$\Rightarrow \frac{k_p}{k_n} = \frac{\gamma \mu_p}{\beta \mu_n} = B^2$$

$$\Rightarrow V_m = \frac{V_{tn} + \sqrt{\frac{\gamma \mu_p}{\beta \mu_n}} \cdot (V_{dd} - |V_{tp}|)}{1 + \sqrt{\frac{\mu_p \cdot \gamma}{\mu_n \cdot \beta}}}$$

Thus, the inverter switching voltage depends on the supply voltage, transistor threshold voltages and relative pmos/nmos strengths.

1.1.4 Read Stability Condition & Cell Ratio Selection

Asserting the expectation that $V_q < V_m$ so that

$$(V_{dd} - V_{tn}) \left(1 - \sqrt{\frac{\beta}{1+\beta}} \right) < \frac{V_{tn} + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}} (V_{dd} - |V_{tp}|)}{1 + \sqrt{\frac{\mu_p \gamma}{\mu_n \beta}}}$$

Increasing β strengthens the pull down transistor relative to the pass gate and reduces the read disturb voltage. However, β also affects the inverter switching voltage and importantly, the area.

Therefore the final cell ratio cannot be selected by minimizing V_q